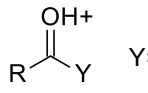
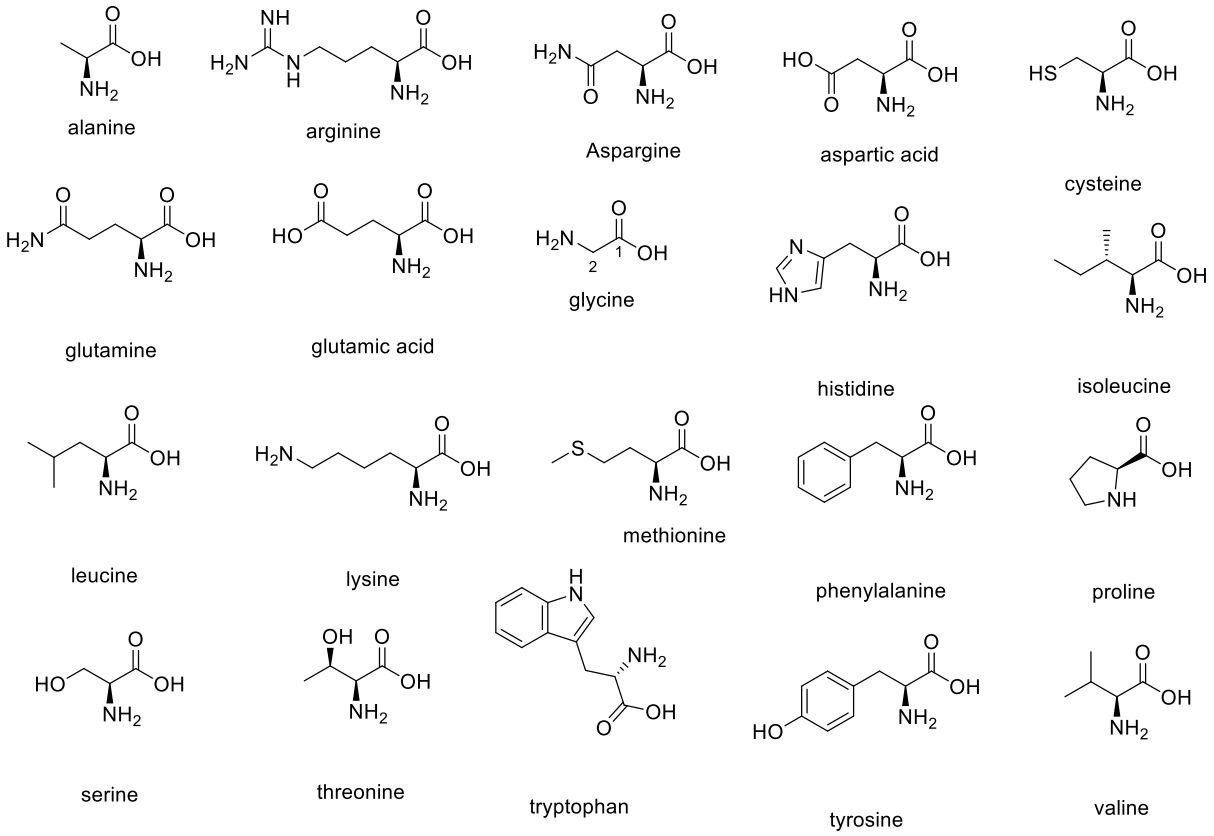
pK_a Table

Acid	Conjugated base	pK_a
H_2SO_4	HSO_4^-	-9
HCl	Cl^-	-6
$ArSO_3H$	$ArSO_3^-$	-6 to -7
 Y=H, R, OR		-6 to-7
H_3O^+	H_2O	-1.74
HF	F^-	3.22
$ArNH_3^+$	$ArNH_2$	5
pyridine·H^+	pyridine	5.2
RCO_2H	RCO_2^-	4-6
H_2CO_3	$HC0_3^-$	6.4
$RC(O)CH_2C(O)R$	$RC(O)CH^-C(O)R$	9
HCO_3^-	CO_3^-	10.33
ArOH	ArO^-	10
RNH_3^+	RNH_2	10
$R_2NH_2^+$	R_2NH	11
R_3NH^+	R_3N	12
$RC(O)CH_2CO_2R$	$RC(O)CH^-CO_2R$	11
$RO_2CCH_2CO_2R$	$RO_2CCH^-CO_2R$	13
H_2O	OH^-	15.74
RCH_2OH	RCH_2O^-	16
R_2CHOH	R_2CHO^-	17
R_3COH	R_3CO^-	18
$RC(O)NH_2$	$RC(O)NH^-$	18-19
$RC(O)CH_2R$	$RC(O)CH^-R$	19-20
RO_2CH_2R	RO_2CH^-R	24
RC≡CH	RC≡C^-	25
NH_3	NH_2^-	33
R_2NH	R_2N^-	35
$R_2C{=}CHCH_3$	$R_2C{=}CHCH_2^-$	43
$R_2C{=}CH_2$	$R_2C{=}CH^-$	44
RH	R^-	50

Amino acid structure



Amino Acids pKa

Amino acid	p*K*a		
	α-CO_2H	α-NH_3^+	Chain
Alanine	2.34	9.69	
Arginine	2.17	9.04	12.48
Aspargine	2.02	8.80	
Aspartic acid	1.88	9.60	3.65
Cystéine	1.96	10.28	8.18
Glutamic Acid	2.19	9.67	4.25
Glutamine	2.17	9.13	
Glycine	2.34	9.60	
Histidine	1.82	9.17	6.00
Isoleucine	2.36	9.60	
Leucine	2.36	9.60	
Lysine	2.18	8.95	10.53
Methionine	2.28	9.21	
Phenylalanine	1.83	9.13	
Proline	1.99	10.60	
Serine	2.21	9.15	

Threonine	2.09	9.10	
Tryptophan	2.83	9.39	
Tyrosine	2.20	9.11	10.07
Valine	2.32	9.62	